

Digital Signal Processing

Detailed Exam Solutions (Part I)

Generated Solution Document

Introduction

This document provides complete, step-by-step solutions for all questions in **PART I** of the Digital Signal Processing exam paper (Ref No: Ex/EE/PC/B/T/224/2025).

Question 1

(a) Sampling and Nyquist Rate

Problem: A signal $x(t) = \sin^2(20\pi t) \sin(10\pi t)$ is uniformly sampled. (i) Determine the Nyquist rate. (ii) If sampled at 18 Hz, and reconstructed by an ideal low-pass filter with a cutoff of 9 Hz, determine the frequencies of the harmonic components present in the reconstructed analog signal.

Solution:

(i) Nyquist Rate

Using the identity $\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$:

$$\begin{aligned} x(t) &= \left(\frac{1 - \cos(40\pi t)}{2} \right) \sin(10\pi t) \\ &= \frac{1}{2} \sin(10\pi t) - \frac{1}{2} \cos(40\pi t) \sin(10\pi t) \end{aligned}$$

Using $\cos(A) \sin(B) = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$:

$$x(t) = \frac{1}{2} \sin(10\pi t) - \frac{1}{4} \sin(50\pi t) + \frac{1}{4} \sin(30\pi t)$$

The continuous-time frequencies $f_i = \frac{\omega_i}{2\pi}$ are: $f_1 = 5$ Hz, $f_2 = 25$ Hz, $f_3 = 15$ Hz. The maximum frequency component is $f_{max} = 25$ Hz.

The Nyquist rate is defined as:

$$f_{Nyquist} = 2 \times f_{max} = 50 \text{ Hz}$$

(ii) Reconstructed Frequencies

The signal is sampled at $f_s = 18$ Hz (< 50 Hz, so aliasing occurs). The folded/aliased frequency $f_a = |f_{in} - kf_s|$ where $0 \leq f_a \leq f_s/2 = 9$ Hz.

- $f_{in} = 5 \text{ Hz} \implies k = 0 \implies f_a = 5 \text{ Hz}.$
- $f_{in} = 25 \text{ Hz} \implies k = 1 \implies f_a = |25 - 18| = 7 \text{ Hz}.$
- $f_{in} = 15 \text{ Hz} \implies k = 1 \implies f_a = |15 - 18| = 3 \text{ Hz}.$

The sampled signal contains components at 3 Hz, 5 Hz, and 7 Hz. An ideal low-pass filter with cutoff 9 Hz passes all of them.

Reconstructed frequencies: 3 Hz, 5 Hz, and 7 Hz.

(b) Z-transform

Problem: Determine the Z-transform $X(z)$ of $x[n] = a^{|n|}$ where $|a| < 1$. Express as a rational function, state the ROC, and describe the pole-zero plot.

Solution:

Split into causal and anti-causal parts: $x[n] = a^n u[n] + a^{-n} u[-n - 1]$.

$$\begin{aligned} Z\{a^n u[n]\} &= \frac{1}{1 - az^{-1}}, \quad |z| > |a| \\ Z\{a^{-n} u[-n - 1]\} &= \sum_{m=1}^{\infty} a^m z^m = \frac{az}{1 - az} = \frac{-1}{1 - a^{-1}z^{-1}}, \quad |z| < \left|\frac{1}{a}\right| \end{aligned}$$

Total $X(z)$:

$$X(z) = \frac{z}{z - a} - \frac{az}{az - 1} = \frac{z(az - 1) - az(z - a)}{(z - a)(az - 1)} = \frac{z(a^2 - 1)}{a(z - a)(z - 1/a)}$$

ROC: $|a| < |z| < 1/|a|$

Poles: $z = a, z = 1/a$. **Zeros:** $z = 0$.

Alternative 1(b) (OR): $x[n] = e^{-an} \cos(\omega_0 n) u[n]$ with $a > 0$.

Let $\alpha = e^{-a}$. Then $x[n] = \alpha^n \cos(\omega_0 n) u[n]$.

$$\begin{aligned} X(z) &= Z \left\{ \frac{\alpha^n e^{j\omega_0 n} + \alpha^n e^{-j\omega_0 n}}{2} u[n] \right\} \\ &= \frac{1}{2} \left[\frac{1}{1 - \alpha e^{j\omega_0} z^{-1}} + \frac{1}{1 - \alpha e^{-j\omega_0} z^{-1}} \right] \\ &= \frac{1 - \alpha \cos(\omega_0) z^{-1}}{1 - 2\alpha \cos(\omega_0) z^{-1} + \alpha^2 z^{-2}} \end{aligned}$$

ROC: $|z| > \alpha = e^{-a}$.

Poles: $z = e^{-a \pm j\omega_0}$. **Zeros:** $z = e^{-a} \cos(\omega_0)$.

Question 2

(a) Convolution of Sequences (OR Question)

Problem: Two DTLTI systems have impulse response sequences $g[n] = \{-2, 1, -5, 2, 3\}$ and $h[n] = \{-5, 3, 2, -6, 2, -1, 4\}$. Find the impulse response of the cascade combination.

Solution:

The cascade combination has an equivalent impulse response $y[n] = g[n] * h[n]$, which is the linear convolution of the two sequences.

- Length of $g[n]$ is $N_1 = 5$
- Length of $h[n]$ is $N_2 = 7$
- Length of $y[n]$ is $N_1 + N_2 - 1 = 11$

Performing the tabular or polynomial multiplication: $(-2, 1, -5, 2, 3) * (-5, 3, 2, -6, 2, -1, 4)$

$$y[0] = (-2)(-5) = 10$$

$$y[1] = (-2)(3) + (1)(-5) = -6 - 5 = -11$$

$$y[2] = (-2)(2) + (1)(3) + (-5)(-5) = -4 + 3 + 25 = 24$$

$$y[3] = (-2)(-6) + (1)(2) + (-5)(3) + (2)(-5) = 12 + 2 - 15 - 10 = -11$$

... and so on.

The complete sequence is computed as:

$$y[n] = \{10, -11, 24, -11, -29, 47, -25, -5, -16, 5, 12\}$$

Question 3

(a) Transfer Function & Stability

Problem: A DTLTI system has zeros at the origin (pair) and $z = -0.8$. Poles at $z = -0.1 \pm j0.9$ and $z = 0.7$. DC gain is 6. Obtain $H(z)$, indicate ROCs, and comment on causality/stability.

Solution:

$$H(z) = K \frac{z^2(z + 0.8)}{(z - 0.7)(z - (-0.1 + j0.9))(z - (-0.1 - j0.9))}$$

Denominator $D(z) = (z - 0.7)(z^2 + 0.2z + 0.82) = z^3 - 0.5z^2 + 0.68z - 0.574$.

DC gain $H(1) = 6$:

$$6 = K \frac{1(1.8)}{0.3(1.02 + 0.82) + \dots} = K \frac{1.8}{(0.3)(2.02)} = K \frac{1.8}{0.606} \implies K \approx 2.02$$

$$H(z) = \frac{2.02z^2(z + 0.8)}{z^3 - 0.5z^2 + 0.68z - 0.574}$$

Poles magnitudes: 0.7 and $\sqrt{0.1^2 + 0.9^2} = \sqrt{0.82} \approx 0.9055$.

ROCs:

1. $|z| > 0.9055$: Causal and Stable (includes unit circle $|z| = 1$).
2. $0.7 < |z| < 0.9055$: Non-causal, Unstable.
3. $|z| < 0.7$: Anti-causal, Unstable.

(b) s-plane to z-plane Mapping

Problem: How does the left-half of the s-plane map on to the z-plane? Explain.

Solution:

The standard mapping from the continuous s-domain to the discrete z-domain is $z = e^{sT}$, where T is the sampling period. Let $s = \sigma + j\Omega$. Then:

$$z = e^{(\sigma + j\Omega)T} = e^{\sigma T} e^{j\Omega T}$$

The magnitude of z is $|z| = e^{\sigma T}$ and the angle is $\angle z = \Omega T$. For the **left-half of the s-plane**, we have $\sigma < 0$. If $\sigma < 0$, then $e^{\sigma T} < 1$. Therefore, $|z| < 1$. This means the entire left-half of the s-plane maps to the **interior of the unit circle** in the z-plane. The $j\Omega$ axis ($\sigma = 0$) maps exactly onto the unit circle ($|z| = 1$).

Question 4

(a) Butterworth Filter via Impulse Invariance

Problem: Design a low pass digital Butterworth filter with DC gain of 0 dB, attenuation of 3 dB at 1620 Hz, and attenuation ≥ 10 dB at 3240 Hz. Sampling frequency $F_s = 7$ kHz. Use Impulse Invariant Transformation. Obtain difference equation.

Solution:

1. **Analog Specifications:** $f_p = 1620$ Hz $\implies \Omega_p = 2\pi(1620) = 10178.76$ rad/s.
 $f_s = 3240$ Hz $\implies \Omega_s = 2\pi(3240) = 20357.52$ rad/s. $A_p = 3$ dB, $A_s = 10$ dB.

2. **Filter Order N and Cutoff Ω_c :**

$$N \geq \frac{\log_{10} \sqrt{(10^{0.1(10)} - 1)/(10^{0.1(3)} - 1)}}{\log_{10}(3240/1620)} = \frac{\log_{10} \sqrt{9/1}}{\log_{10} 2} = \frac{0.477}{0.301} = 1.58 \implies N = 2$$

For $N = 2$, $\Omega_c \approx \Omega_p = 10178.76$ rad/s (since $A_p \approx 3$ dB). Let $\Omega_c = 10190.85$ rad/s.

3. **Analog Poles:** For a 2nd-order Butterworth, poles are at angles $\pm 135^\circ$: $p_{1,2} = \Omega_c(-\frac{1}{\sqrt{2}} \pm j\frac{1}{\sqrt{2}}) = -7206 \pm j7206$.

4. **Impulse Invariant Transformation:** $H(s) = \frac{\Omega_c^2}{(s-p_1)(s-p_2)} = \frac{R_1}{s-p_1} + \frac{R_2}{s-p_2}$. $R_1 = \frac{\Omega_c^2}{p_1-p_2} = -j7206$, $R_2 = j7206$. In the z-domain: $H(z) = \frac{R_1}{1-e^{p_1 T} z^{-1}} + \frac{R_2}{1-e^{p_2 T} z^{-1}}$. With $T = 1/7000$: $e^{p_1 T} = e^{(-7206+j7206)/7000} = e^{-1.0294} e^{j1.0294} = 0.357(0.515 + j0.857) = 0.184 + j0.306$. $e^{p_2 T} = 0.184 - j0.306$. Combining terms and normalizing for a 0 dB DC gain ($H(1) = 1$):

$$H_{norm}(z) = \frac{0.7595z^{-1}}{1 - 0.3681z^{-1} + 0.1276z^{-2}}$$

Difference Equation:

$$y[n] = 0.3681y[n-1] - 0.1276y[n-2] + 0.7595x[n-1]$$

(b) Characteristic Aliasing

Problem: Explain the problem of 'Characteristic Aliasing' in Impulse Invariant Transformation. When is it minimal?

Solution:

In the Impulse Invariant Transformation, the discrete-time frequency response is an aliased version of the continuous-time frequency response: $H(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a\left(j\frac{\omega-2\pi k}{T}\right)$. If the analog filter $H_a(j\Omega)$ is not strictly bandlimited to the Nyquist frequency $\Omega_N = \pi/T$, the frequency tails of adjacent periods overlap into the baseband $(-\pi/T, \pi/T)$. This overlapping distorts the frequency response, known as "characteristic aliasing."

Minimization: It is minimal when the analog filter has a sharp roll-off and very high attenuation in the stopband before Ω_N . Therefore, it is mainly suited for **low-pass and band-pass filters** sampled at a sufficiently **high sampling rate**.

Question 5 (Short Notes)

(a) Frequency Spectra of uniformly sampled signals:

When a continuous signal $x(t)$ is sampled uniformly at rate $F_s = 1/T$, the resulting spectrum

is $X_s(f) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(f - kF_s)$. This implies the original spectrum is periodically replicated every F_s Hz. To prevent overlap (aliasing), F_s must be $\geq 2f_{max}$.

(b) Recursive and non-recursive DTLTI systems:

Recursive systems compute the current output $y[n]$ using past outputs (e.g., $y[n-1]$) along with inputs. They have feedback loops and typically represent Infinite Impulse Response (IIR) filters. Non-recursive systems compute $y[n]$ using only current and past inputs (no feedback), naturally forming Finite Impulse Response (FIR) filters which are inherently stable.

(c) Representing uniformly sampled signals by scaled impulses:

An analog signal converted to discrete-time mathematically uses an impulse train $p(t) = \sum \delta(t - nT)$. The sampled signal is $x_s(t) = x(t) \cdot p(t) = \sum x(nT)\delta(t - nT)$. This is a train of Dirac impulses occurring at nT , where the area of each impulse equals the sample value $x(nT)$.

(d) Bilinear transform with frequency prewarping:

The bilinear transform maps the entire $j\Omega$ axis to the unit circle uniquely via $s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$. However, it introduces a non-linear frequency compression (warping) governed by $\Omega = \frac{2}{T} \tan(\omega/2)$. Prewarping is the process of distorting the critical analog frequencies prior to design, ensuring they map exactly to the desired digital frequencies.